

Md Enamul Islam Bhuiyan Meraj

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RESEARCH INTERESTS

Geotechnical Earthquake Engineering; Liquefaction Hazard Assessment & Mitigation; Seismic Microzonation & Geospatial Analysis; Probabilistic Risk & Reliability Assessment; Computational Geotechnics & Soil–Structure Interaction.

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

Dhaka, Bangladesh

M.Sc. Engg. (Civil & Geotechnical) | CGPA 3.50/4.00

May 2025 – Jan 2027 (expected)

- **Thesis:** *Shear Wave Velocity Based Deterministic and Probabilistic Liquefaction Assessment of DMDP Area in Bangladesh.*
Supervisor: Prof. Dr. Mehedi Ahmed Ansary.
- Completed graduate coursework: Soil Mechanics I and II, Earthquake Engineering, Reinforced Earth, Constitutive Modeling in Soil Mechanics, Earth Dams and Stability of Slopes.

Rajshahi University of Engineering and Technology (RUET)

Rajshahi, Bangladesh

B.Sc. in Civil Engineering | CGPA 3.63/4.00

Feb 2019 – Apr 2024

- **Thesis:** *Behavior of Single Pile in Cohesionless Soil on Horizontal and Sloping Ground Surface under Lateral Loading.*
Supervisor: Prof. Dr. Md. Abdul Alim.

RESEARCH EXPERIENCE

Postgraduate Fellow

Dec 2025 – Present

Department of Civil Engineering, BUET. Supervisor: Prof. Dr. Mehedi Ahmed Ansary.

Merit-based BUET Postgraduate Fellowship with a 40-hour/week research commitment under faculty supervision.

Regional V_s -Based Liquefaction Assessment of the DMDP Area:

- The assessment is being applied to a regional dataset of approximately 400 downhole V_s profiles with associated SPT and geotechnical borehole data, spanning Pleistocene and Holocene deposits across the 1,530 km² DMDP area.
- Following susceptibility screening, deterministic triggering is being assessed using the Andrus–Stokoe and Kayen et al. (2013) procedures; a Monte Carlo framework is being developed for uncertainty analysis, with the Kayen probabilistic model and Juang Bayesian mapping used for comparison. Liquefaction manifestation is being evaluated using LPI, LPI_{ISH}, and LSN, followed by GIS-based regional mapping of liquefaction potential and manifestation severity.

Systematic Review of Vacuum-Based Soft-Ground Improvement with Emphasis on HVDM:

- Contributed to a PRISMA-guided systematic review and manuscript drafting on vacuum-based soft-ground improvement, synthesizing 69 eligible studies from 860 records and comparing treatment mechanisms, geotechnical performance, field applicability, and research gaps, with emphasis on HVDM and a method-selection framework.

Comparative LCA and LCCA of Geotextile-Reinforced and Conventional Road Embankments:

- Developing a comparative LCA and LCCA of a geotextile-reinforced wrapped-face road embankment and a conventional 1:1 earthen slope using openLCA (ecoinvent v3.12, EF 3.1), considering construction, transportation, water-induced erosion and reconstruction, maintenance, long-term cost, and sensitivity analysis.

Effect of Sand Gradation and TDA Content on Shear Strength and Compressibility of Sand–TDA Mixtures:

- Contributing to an experimental study through literature review, research-objective development, direct shear testing, and ongoing consolidation testing to evaluate the effects of sand gradation and tire-derived aggregate (TDA) content on shear strength and compressibility; findings are intended for journal publication.

Post-Earthquake Reconnaissance Survey

Narsingdi & Narayanganj | Nov 2025

BUET research team response to the 21 November 2025 Madhabdi earthquake.

- Conducted ATC-20 rapid safety evaluations of more than 50 structures through exterior and interior inspections; recorded construction type, structural damage and hazards, applicable ATC-20 safety category, GPS location, occupancy, and available foundation information.

Undergraduate Thesis Research

Rajshahi, Bangladesh | 2023 – 2024

Geotechnical Engineering Laboratory, RUET. Supervisor: Prof. Dr. Md. Abdul Alim.

Lateral Response of Single Piles on Level and Sloping Ground:

- Conducted model-scale lateral load tests on 25-mm stainless-steel piles in cohesionless soil under level, 1V:6H, and 1V:3H ground conditions ($L/D = 23$ and 26), with loading along and against the slope; results showed lower lateral resistance on sloping ground, greater reduction for the steeper slope, and higher resistance with increasing L/D ratio.

PUBLICATIONS & MANUSCRIPTS

Journal Manuscript Under Peer Review

- Mottaqi, A., **Meraj, M. E. I. B.**, & Ansary, M. A. Vacuum-based soft soil improvement: A PRISMA-guided systematic review with special emphasis on the High Vacuum Densification Method (HVDM). Geotechnical and Geological Engineering. Submitted July 2026.

Conference Proceedings

- **Meraj, M. E. I. B.**, Hridoy, K. H., & Alim, M. A. (2024). Behavior of single pile in cohesionless soil on horizontal and sloping ground surface under lateral loading. Proceedings of the 7th International Conference on Advances in Civil Engineering (ICACE 2024), CUET, Chattogram, Bangladesh.

Manuscripts in Preparation

- **Meraj, M. E. I. B.**, & Ansary, M. A. Regional liquefaction assessment of the DMDP area, Bangladesh, using shear wave velocity profiles: Deterministic, probabilistic, and manifestation-based approaches.
- **Meraj, M. E. I. B.**, & Ansary, M. A. Comparative life cycle assessment and life cycle cost assessment of a geotextile-reinforced wrapped-face road embankment and a conventional earthen slope.

NUMERICAL MODELING

Self-directed PLAXIS 2D/3D studies based on published papers, with key model outputs compared with reported results.

- **Deep Excavations:** Modeled the Bangkok MRT Sukhumvit Station, central Hanoi, and Ho Chi Minh City excavation problems. Used staged excavation to obtain diaphragm-wall deflection and ground-surface settlement; for the Ho Chi Minh City model, also examined adjacent-building settlement.
- **TBM Tunneling:** Modeled a greenfield TBM tunneling problem in layered soil and obtained the surface-settlement profile above the tunnel, comparing the PLAXIS results with the published study.
- **Soft-Ground Embankments & Ground Improvement:** Modeled a DCM-column-supported embankment with a fiber-reinforced load-transfer platform and obtained settlement, lateral displacement, and stress-transfer results. Also developed staged road-embankment and PVD-assisted consolidation models.

TECHNICAL SKILLS

- **Software:** PLAXIS 2D/3D (finite-element modeling), GeoStudio (SEEP/W; seepage analysis), ArcGIS Pro (hazard mapping), openLCA, AutoCAD
- **Laboratory and field:** Geotechnical laboratory testing (index and engineering properties), direct shear and consolidation testing, electrical resistivity tomography (electrode layout and data acquisition, assisted), ATC-20 rapid safety evaluation.
- **Programming:** Python (Monte Carlo simulation)

AWARDS AND FELLOWSHIPS

- Postgraduate Fellowship, Bangladesh University of Engineering and Technology 2025
- Technical Education Scholarship, Bangladesh Technical Education Board 2019 – 2022
- Board Scholarship, Dhaka Education Board (Secondary School Certificate) 2016